CampusLaundry Management System

Data Manipulation Language (DML) Specification & Operational Queries

Project: CampusLaundry Management System	**Category:** DML Commands & Query Operations	**DBMS Engine:** MySQL 8.0+ / InnoDB
Schema Normalization: 3NF (Strict Third Normal Form)	**Academic Scope:** Database Management Systems Laboratory	**Deployment:** Cloud Production Verified (Aiven)

1. DML Architecture & Transactional Lifecycle Overview

Data Manipulation Language (DML) statements manage data ingestion, student lifecycle updates, order state transitions, and analytical reporting across the CampusLaundry ecosystem. All operational transactions strictly enforce referential integrity and atomic quota balancing.

2. Master Catalog & Entity Data Ingestion

2.1 Vendor Registration (Contractors)

```
INSERT INTO vendor (Vendor_ID, Vendor_Name, Contact_No) VALUES
('VND01', 'Sparkle Wash Services', 9876543210),
('VND02', 'Campus Cleaners Ltd.', 9123456789);
```

2.2 Drop Point Kiosk Initialization

```
INSERT INTO drop_point (Drop_Point_ID, Location, Operational_Status, Vendor_ID) VALUES
('DP01', 'B1 Ground Floor Locker', 'Active', 'VND01'),
('DP02', 'B11 Laundry Hall', 'Active', 'VND01'),
('DP03', 'G1 Reception Wing', 'Active', 'VND02'),
('DP04', 'G5 Basement Facility', 'Active', 'VND02');
```

2.3 Garment Catalog with Weight Benchmarks (kg)

```
INSERT INTO laundry_item (Item_ID, Item_Name, Category, Standard_Weight) VALUES
('ITM01', 'T-Shirt / Shirt', 'Daily Wear', 0.25),
('ITM02', 'Jeans / Trousers', 'Daily Wear', 0.65),
('ITM03', 'Bed Sheet (Single)', 'Bedding', 0.60),
('ITM04', 'Bed Sheet (Double)', 'Bedding', 1.10),
('ITM05', 'Bath Towel', 'Daily Wear', 0.45),
('ITM06', 'Pillow Cover', 'Bedding', 0.15),
('ITM07', 'Sweater / Hoodie', 'Woolen', 0.80),
('ITM08', 'Blanket / Comforter', 'Bedding', 2.20),
('ITM09', 'Shorts / Lower', 'Daily Wear', 0.30),
('ITM10', 'Lab Coat / Uniform', 'Formals', 0.40);
```

2.4 Student Cohort Profile Ingestion

```
INSERT INTO students
(std_id, std_name, contact, hostel, room, academic_year, quota, cycle_used, cycle_remaining)
VALUES
('2402050088', 'Ananya Iyer', 9870044556, 'G1', '405', '3rd Year', 30, 29, 1),
('2502050112', 'Priya Patel', 9870022334, 'G8', '108', '2nd Year', 30, 12, 18),
('2502050650', 'Arush Bhalu', 9666666666, 'G4', '699', '2nd Year', 15, 0, 15),
('2502050680', 'Aarav Sharma', 9870011223, 'B10', '205', '2nd Year', 30, 6, 24),
('2602050045', 'Rohan Verma', 9870033445, 'B4', '312', '1st Year', 30, 2, 28);
```

2.5 Administrative Staff Seed Accounts (bcrypt hash)

```
INSERT INTO admin_users (username, password_hash, full_name, role, is_active) VALUES
('warden_b1', '$2b$12$e8K7bAOmf6wz7f1P0p9vEOZk5U6k8x2rJ1v8m6x5n3p2q1w4e5r6t', 'Col. R. K. Singh', 'warden', TRUE),
('vendor_sparkle', '$2b$12$e8K7bAOmf6wz7f1P0p9vEOZk5U6k8x2rJ1v8m6x5n3p2q1w4e5r6t', 'Sparkle Wash Manager', 'vendor', TRUE),
('superadmin', '$2b$12$e8K7bAOmf6wz7f1P0p9vEOZk5U6k8x2rJ1v8m6x5n3p2q1w4e5r6t', 'Campus Facilities Director', 'superadmin', TRUE);
```

3. Operational Transaction Logging (3NF Compliant)

Demonstrates dropping off laundry and registering line items. Vendor_ID is omitted; vendor attribution is handled by joining with drop_point.

```
-- Step 1: Create Laundry Batch Header
INSERT INTO laundry_transaction (
    Transaction_ID, std_id, Drop_Point_ID, Drop_Off_Date,
    Expected_Completion_Date, Total_Weight, Cycles_Deducted, Status
) VALUES (
    'TXN1002', '2502050680', 'DP01', CURRENT_DATE(),
    DATE_ADD(CURRENT_DATE(), INTERVAL 3 DAY), 4.85, 1, 'Submitted'
);

-- Step 2: Ingest Itemized Garment Manifest
INSERT INTO transaction_item (Transaction_ID, Item_ID, Quantity) VALUES
('TXN1002', 'ITM01', 6),  -- 6 T-Shirts (1.50 kg)
('TXN1002', 'ITM02', 2),  -- 2 Jeans (1.30 kg)
('TXN1002', 'ITM03', 1),  -- 1 Single Bed Sheet (0.60 kg)
('TXN1002', 'ITM05', 2),  -- 2 Bath Towels (0.90 kg)
('TXN1002', 'ITM09', 1);  -- 1 Shorts (0.30 kg)

-- Step 3: Deduct Cycle Quota from Student Ledger
UPDATE students
SET cycle_used = cycle_used + 1,
    cycle_remaining = cycle_remaining - 1
WHERE std_id = '2502050680' AND cycle_remaining >= 1;
```

4. Lifecycle State Updates & Quota Refunds

State transitions across the cleaning pipeline. Changing status to 'Completed' triggers completion timestamp capture.

```
-- 1. Advance Batch to Washing Stage
UPDATE laundry_transaction
SET Status = 'In Washing'
WHERE Transaction_ID = 'TXN1002';

-- 2. Stage to Ready for Pickup (Trigger sets Actual_Completion_Date)
UPDATE laundry_transaction
SET Status = 'Ready for Pickup'
WHERE Transaction_ID = 'TXN1002';

-- 3. Finalize Order Pickup by Student
UPDATE laundry_transaction
SET Status = 'Completed'
WHERE Transaction_ID = 'TXN1002';

-- 4. Order Cancellation with Automated Quota Restoration
UPDATE laundry_transaction
SET Status = 'Cancelled'
WHERE Transaction_ID = 'TXN1002';
```

5. Critical Relational Retrieval & Analytics Queries

5.1 3NF Traversal: Order History with Kiosk & Vendor Resolution

```
SELECT
    t.Transaction_ID,
    t.std_id,
    s.std_name,
    s.hostel,
    dp.Location AS drop_point_location,
    v.Vendor_Name,
    t.Drop_Off_Date,
    t.Expected_Completion_Date,
    t.Total_Weight,
    t.Status
FROM laundry_transaction t
JOIN students s ON t.std_id = s.std_id
JOIN drop_point dp ON t.Drop_Point_ID = dp.Drop_Point_ID
JOIN vendor v ON dp.Vendor_ID = v.Vendor_ID
WHERE t.std_id = '2502050680'
ORDER BY t.Drop_Off_Date DESC;
```

5.2 Campus-Wide Kiosk Load Distribution

```
SELECT
    dp.Drop_Point_ID,
    dp.Location,
    v.Vendor_Name,
    COUNT(t.Transaction_ID) AS active_batches,
    COALESCE(SUM(t.Total_Weight), 0.00) AS total_pending_kg
FROM drop_point dp
JOIN vendor v ON dp.Vendor_ID = v.Vendor_ID
LEFT JOIN laundry_transaction t
    ON dp.Drop_Point_ID = t.Drop_Point_ID
    AND t.Status NOT IN ('Completed', 'Cancelled')
GROUP BY dp.Drop_Point_ID, dp.Location, v.Vendor_Name
ORDER BY active_batches DESC;
```

5.3 Quota Risk Alert Query (Students with <= 3 Cycles)

```
SELECT
    std_id,
    std_name,
    hostel,
    room,
    quota,
    cycle_used,
    cycle_remaining,
    ROUND((cycle_used / quota) * 100, 1) AS burn_pct
FROM students
WHERE cycle_remaining <= 3
ORDER BY cycle_remaining ASC, burn_pct DESC;
```

6. Purge & Deletion Operations

Demonstrates safe deletion complying with ON DELETE CASCADE rules.

```
-- Cascading deletion: removes transaction header and all child transaction_items automatically
DELETE FROM laundry_transaction
WHERE Transaction_ID = 'TXN1002' AND Status = 'Cancelled';
```